



MRS6130-AIP-DEV-V1

mmWave Radar Development Board

User Guide

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Revision History

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1.4	Update section 2.2 component introduction: Resistance.	2025-09-23
1.3	Add a new section 2.2, "Component introduction: Resistance".	2025-09-13
1.2	Change to the content related to the mrs6130-AIP-DEV-V1 development board.	2025-05-19
1.1	Modify the description of the power options and power consumption measurement methods.	2024-08-20
1.0	Initial release.	2024-04-20

Table of Contents

Revision History.....	ii
Table of Contents.....	iii
List of Figures.....	iv
Table of Contents.....	v
1 Overview.....	1
2 Hardware Introduction.....	2
2.1 Hardware block diagram.....	2
2.2 Component introduction.....	2
2.3 Power options.....	5
2.4 Pin header.....	5
3 Hardware Usage.....	7
3.1 Required hardware.....	7
3.2 Module connection.....	7
3.3 Basic usage method.....	7
3.4 Power consumption measurement method.....	7

List of Figures

Figure 2-1	Schematic Block Diagram of MRS6130-AIP-DEV-V1 Hardware Connection.....	2
Figure 2-2	Physical Image of Development Board Components.....	2
Figure 3-1	P1806.....	7
Figure 3-2	P1812.....	7
Figure 3-3	Power measurement instrument and hardware configuration.....	8

Table of Contents

Table 2-1	Description of Development Board Components.....	3
Table 2-2	Definition and Description of Pin Header Signals.....	5

1 Overview

The RS6130-AIP-DEV-BOX development kit is an application development platform launched by Possumic Technology, integrating ultra-low power 60GHz FMCW millimeter-wave radar, Bluetooth 5 (LE), and IEEE 802.15.4 wireless functions into an intelligent radar sensing chip. It is designed to facilitate developers in scheme evaluation and design. This development kit includes the MRS6130 series modules (MRS6130-P1806, MRS6130-P1812 modules) and an MRS6130-AIP-DEV-V1 development board base, allowing developers to connect a wide range of peripherals to meet the development needs of various application scenarios. It is particularly suitable for applications such as smart home, smart appliances, smart security, and smart lighting, and its ultra-low power consumption feature can provide longer battery life for battery-powered products.

This document primarily introduces the usage of the MRS6130-AIP-DEV-V1 development board baseboard, encompassing component information, power supply information, interface information, and other pertinent details of this evaluation baseboard.

2 Hardware Introduction

2.1 Hardware block diagram

The main components and application connection methods of MRS6130-AIP-DEV-V1 are shown in the figure below.

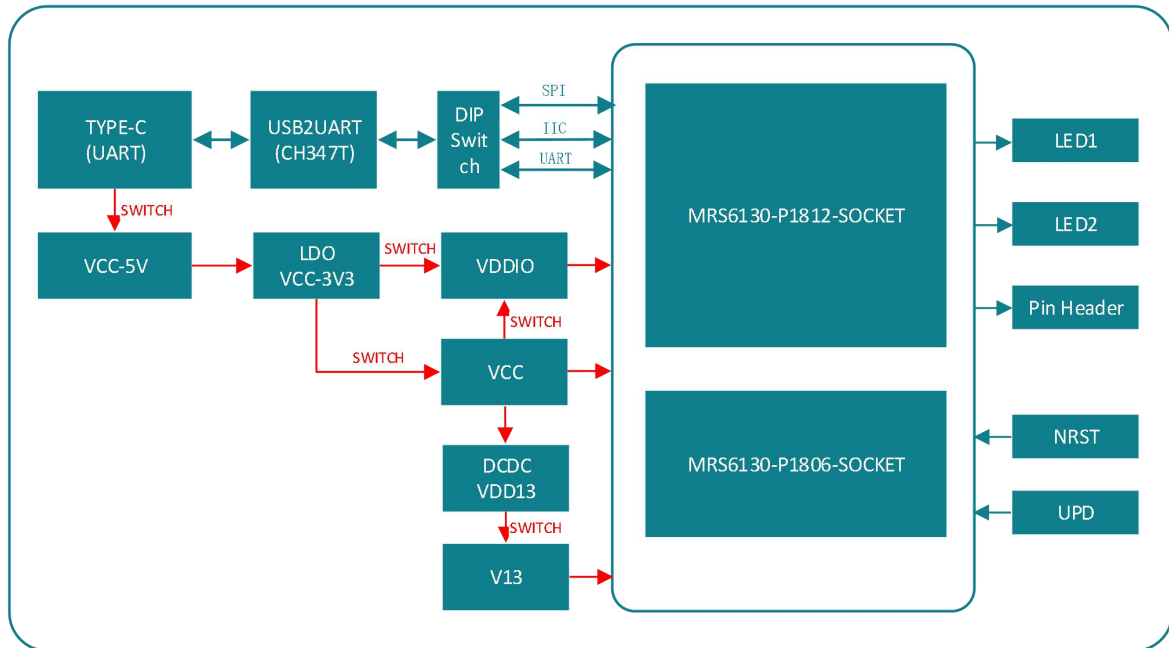


Figure 2-1 Schematic Block Diagram of MRS6130-AIP-DEV-V1 Hardware Connection

2.2 Component introduction

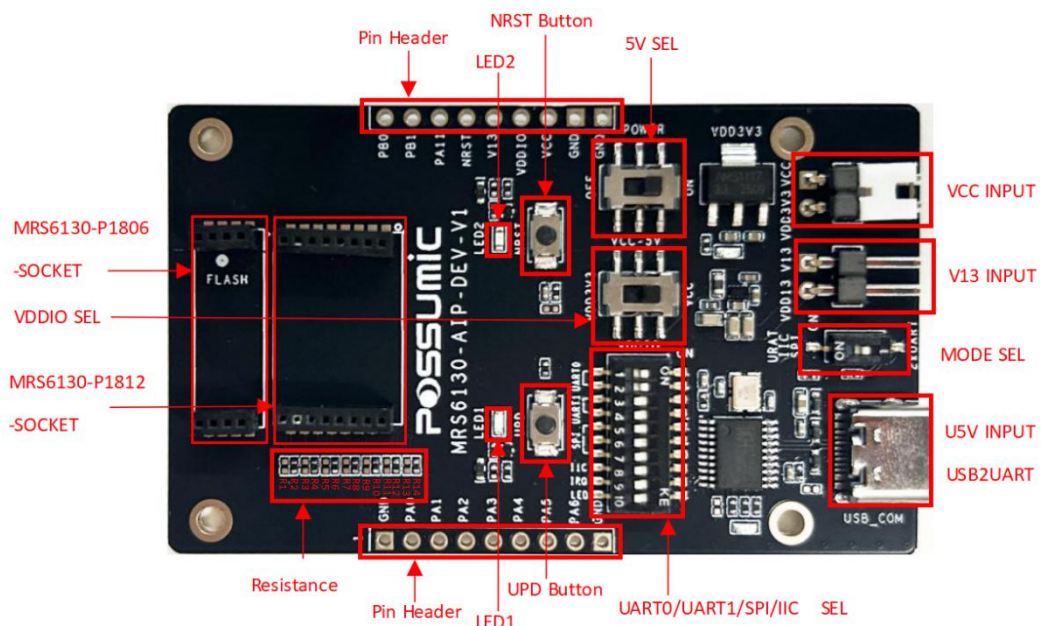


Figure 2-2 Physical Image of Development Board Components

The following describes the components on the development board.

Table 2-1 Description of Development Board Components

Component		Instructions
Module Interface	MRS6130-P1812-SOCKET	Used to connecting the millimeter-wave radar module MRS6130-P1812
	MRS6130-P1806-SOCKET	Used to connecting the millimeter-wave radar module MRS6130-P1806
Button	UPD Button	Upgrade download button In the power-off state, first press the UPD Button, then plug in the USB cable to power on. The chip will start up and enter "Firmware Download" mode, and then download the firmware through the serial port UART0.
	NRST Button	Module reset button
Power Supply	U5V INPUT	Type-C VBUS 5V power supply, and also serves as a UART to USB conversion path, connected to the USB port of a PC.
	5V SEL	Switch for the main power supply VCC-5V of the development board: ➤ Turn to "ON" (default): Turn on the VCC-5V power supply. ➤ Turn to "OFF": Turn off the VCC-5V power supply.
	VDDIO SEL	Selection of the source for module power supply VDDIO, which is used to power the module IO and peripherals: ➤ Set to "VCC" (default): Use VCC as the power supply ➤ Set to "VDD3V3": Use VDD3V3 as the power supply
	VCC INPUT	Module VCC power supply switch, VCC is the main power supply for the radar module: ➤ Connect the jumper cap (default): Use VDD3V3 supplied by the LDO output. The LDO is a component on the development board, and its input power supply is VCC-5V. ➤ Disconnect the jumper cap: If an external DC power supply is required, remove the jumper cap and connect the external power supply to the VCC pin of the Pin Header.
	V13 INPUT	Power supply selection for module V13, where V13 serves as the main power source for the RF part of the radar module: ➤ Connect the jumper cap (default): Use VDD13 supplied by the DCDC output. The DCDC is a component on the development board, and its input power supply is VCC. ➤ Disconnect the jumper cap: Without using VDD13 for power supply, the radar module will adaptively use the internal LDO. At this time, the V13 network voltage is 1.5V.

Component		Instructions
Other	USB2UART	Type-C USB interface, which is used for power supply and UART/IIC/SPI communication of the development board. When the MODE SEL is set to OFF, the USB interface supports dual serial ports. After connecting to a PC, the device name is identified as "USB Serial Device COM*" (where * represents the serial port number, with the larger number corresponding to the radar's UART0 and the smaller number corresponding to the radar's UART1). When the MODE SEL is set to ON, the USB interface supports single serial port/IIC/SPI simultaneously. After connecting to a PC, the device name is recognized as "USB Serial Device COM*".
	LED	Apply indicator LED lights. The power supply comes from VCC-3V3. The control signal for LED1 is PA6, and the control signal for LED2 is PA11.
	MODE SEL	CH347T working mode selection (dip switch): ➤ ON (default): USB to high-speed single serial port + USB to 2-wire IIC + USB to 4-wire SPI ➤ OFF: USB to high-speed dual serial ports Note: The direction of ON is subject to the mark on the dial switch
	UART0/UART1/SPI/IIC SEL	MRS6130 communication mode selection (dip switch): ➤ Set 1 and 2 to ON, and set the others to OFF: UART0 is connected to Type-C ➤ Set pins 3 and 4 to ON, and all other pins to OFF: UART1 is connected to Type-C ➤ Set pins 5, 6, and 7 to ON, and all other pins to OFF: SPI is connected to Type-C ➤ 8 is set to ON, and the others are set to OFF: IIC is connected to Type-C ➤ 9 is set to ON, 10 is set to OFF: PA6 is used to report interrupt data to the USB chip ➤ 10 is set to ON, 9 is set to OFF: PA6 is used to light up LED1
	Pin Header	All available IO pins of the module (excluding the SPI interface of Flash) are brought out to the pin header holes of the development board, including power supplies VCC, VDDIO, and V13.

Component		Instructions
	Resistance	If UART1 (PA0, PA1) needs to be used on the pin header, OR resistors need to be soldered on R11 and R9. If SPI (PA0, PA1, PA2, PA3) is required to be used on the pin header, OR resistors need to be soldered on R11, R9, R7, and R5. If IIC (PA4, PA5) is required to be used on the pin header, OR resistors need to be soldered on R3 and R1. When using the 1806 module, if reporting is required, solder R13; if lighting is needed, solder R14. When using the 1812 module, resistors cannot be soldered onto R13 and R14 simultaneously; both resistor positions can remain vacant.

2.3 Power options

1. Development board power supply:

1) USB_COM interface power supply: Set the DIP Switch VCC-5V to ON (default);

2. There are two options for supplying power to the module VCC:

1) Development board VDD3V3 power supply: VCC INPUT is short-circuited, meaning VDD3V3 and VCC (default);

2) External power supply: Power is supplied through the VCC pin of the Pin Header, with the VCC INPUT disconnected.

3. There are two options for VDDIO power supply of development boards and modules:

1) Development board VDD3V3 power supply: Set the dip switch VDDIO to VDD3V3 (default);

2) VCC power supply: Set the dip switch VDDIO to VCC.

4. There are two power supply options for module V13:

1) VINT power supply output by the module: V13 INPUT is disconnected (default);

2) Development board DCDC power supply: V13 INPUT is short-circuited.

2.4 Pin header

The names and functions of the pin headers on both sides of the development board are listed as follows:

Table 2-2 Definition and Description of Pin Header Signals

Serial Number	Name	Type	Function
1	GND	G	Ground Pin, Short directly to board ground plane.
2	PA0	IO	PA0, WIO0, SPI-CS, UART1-TX, PWM0, MCO
3	PA1	IO	PA1, WIO1, SPC-CLK, UART1-RX, PWM1
4	PA2	IO	PA2, WIO2, SPI-MISO, UART1-RTS, IIC0-SDA, UART2-TX, PWM2
5	PA3	IO	PA3, WIO3, SPI-MOSI, UART1-CTS, IIC0-SCL, UART2-RX, PWM3
6	PA4	IO	PA4, WIO4, SPI-HOLD, IIC0-SDA, PWM4
7	PA5	IO	PA5, WIO5, SPI-WP, IIC0-SCL, PWM5
8	PA6	IO	PA6, WIO6, UART1-RTS, UART2-TX, IIC1-SDA, PWM6, GPADC6
9	GND	G	Ground Pin, Short directly to board ground plane.

Serial Number	Name	Type	Function
10	GND	G	Ground Pin, Short directly to board ground plane.
11	GND	G	Ground Pin, Short directly to board ground plane.
12	VCC	PI	Development board VCC power supply
13	VDDIO	PI	Development board powered by VDDIO
14	V13	PI/PO	The development board V13 supplies power. When there is no external DCDC, V13 outputs 1.5V; when there is an external DCDC, the input is 1.3V
15	NRST	I	Module reset signal
16	PA11	IO	PA11, WIO11, IIC1-SCL, PWM3, GPADC3
17	PB1	IO	PB1, WIO17, UART0-RX, PWM1
18	PB0	IO	PB0, WIO16, UART0-TX, PWM0

3 Hardware Usage

3.1 Required hardware

Hardware preparation for using the MRS6130-AIP-DEV-V1 development board:

- 1) Module MRS6130-P1806/MRS6130-P1812;
- 2) Type-C USB data cable;
- 3) Computer, and install the CH347T serial port driver;

3.2 Module connection

The radar module has two specifications, namely MRS6130-P1806 and MRS6130-P1812. The connection method is shown in the figure below.



Figure 3-1 P1806



Figure 3-2 P1812

3.3 Basic usage method

- **Step 1:** Configure the development board power supply according to the default connection method described in Table 2-1 Description of Development Board Components [Table 2-1 Description of Development Board Components](#).
- **Step 2:** Configure the signal path according to the default connection methods described in [Table 2-1, specifically for MODE SEL](#) and UART0/UART1/SPI/IIC SEL in the development board component instructions.
- **Step 3:** Insert the radar module MRS6130-P1806 or MRS6130-P1812, ensuring to identify the orientation of the module's PIN1. Both the module and the development board have corresponding white dot markers.
- **Step 4:** Connect the PC to the U5V TYPE-C port of the development board via a Type-C data cable. Two serial port devices will appear on the computer side. The serial port with the larger number corresponds to the UART0 of the radar, while the serial port with the smaller number corresponds to the UART1 of the radar.
- **Step 5:** Operate using the PC host computer tool.

3.4 Power consumption measurement method

The development board supports power consumption measurement for evaluation modules, encompassing both scenarios without DCDC and with externally plugged-in DCDC.

- **Step 1:** Set "5V SEL" to "ON", set "VDDIO SEL" to "VCC", and disconnect the jumper cap of "VCC INPUT".

- **Step 2:** The MRS6130-P1806 module does not come with a DCDC module. If you need to measure the operating conditions with a DCDC, connect the jumper cap of "V13 INPUT"; if you need to measure the operating conditions without a DCDC, disconnect the jumper cap of "V13 INPUT".

The MRS6130-P1812 module comes with a built-in DCDC module. When measuring, disconnect the jumper cap for "V13 INPUT".

- **Step 3:** Use a TYPE-C data cable to supply power to the U5V TYPE-C port.
- **Step 4:** Use an external high-precision power measurement instrument to supply power to VCC and GND of the PIN Header, and record the current data.

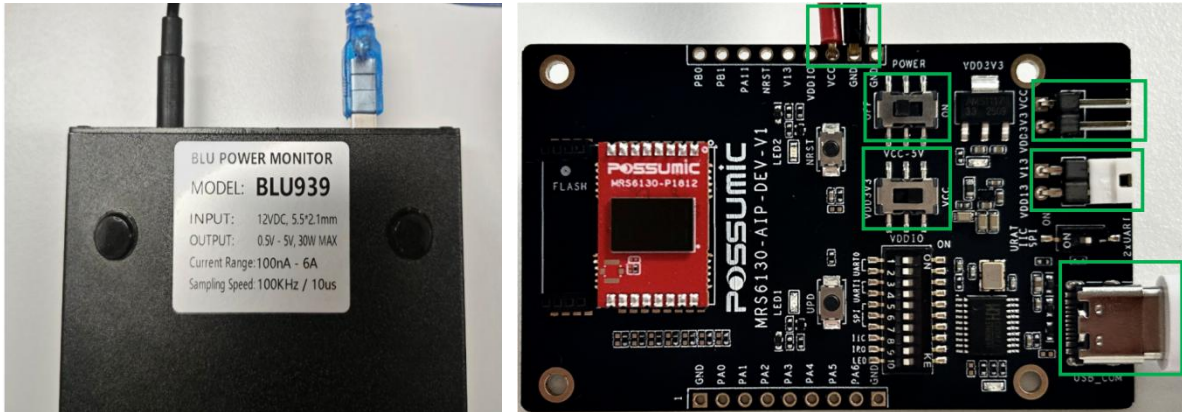


Figure 3-3 Power measurement instrument and hardware configuration

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